

$$\begin{array}{c}
\begin{array}{cc}
\mathcal{K}_{0^+}^{\#1} & \mathcal{K}_{0^+}^{\#1\alpha\beta\chi} \\
\mathcal{K}_{0^+}^{\#1} \dagger & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} \\
\mathcal{K}_{0^+}^{\#1} \dagger^{\alpha\beta\chi} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array}
\end{array}
\begin{array}{cccc}
\mathcal{K}_{1^+}^{\#1\alpha\beta} & \mathcal{K}_{1^+}^{\#2\alpha\beta} & \mathcal{K}_{1^+}^{\#1\alpha} & \mathcal{K}_{1^+}^{\#2\alpha} \\
\mathcal{K}_{1^+}^{\#1} \dagger^{\alpha\beta} & \begin{array}{|c|c|} \hline Y_1 k^2 & \frac{Y_1 k^2}{\sqrt{2}} \\ \hline \end{array} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} \\
\mathcal{K}_{1^+}^{\#2} \dagger^{\alpha\beta} & \begin{array}{|c|c|} \hline \frac{Y_1 k^2}{\sqrt{2}} & \frac{Y_1 k^2}{2} \\ \hline \end{array} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} \\
\mathcal{K}_{1^+}^{\#1} \dagger^\alpha & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} & \begin{array}{|c|c|} \hline \frac{Y_2 k^2}{2} & -\frac{Y_3 k^2}{2\sqrt{2}} \\ \hline \end{array} & \begin{array}{|c|c|} \hline -\frac{Y_3 k^2}{2\sqrt{2}} & \frac{Y_4 k^2}{2} \\ \hline \end{array} \\
\mathcal{K}_{1^+}^{\#2} \dagger^\alpha & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} & \begin{array}{|c|c|} \hline -\frac{Y_3 k^2}{2\sqrt{2}} & \frac{Y_4 k^2}{2} \\ \hline \end{array} & \begin{array}{|c|c|} \hline \mathcal{K}_{2^+}^{\#1\alpha\beta} & \mathcal{K}_{2^+}^{\#1\alpha\beta\chi} \\ \hline \end{array}
\end{array}
\begin{array}{cc}
\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} \\
\mathcal{K}_{2^+}^{\#1} \dagger^{\alpha\beta\chi} & \begin{array}{|c|c|} \hline 0 & \frac{3k^2 \mathcal{K}_{15}^{(4)}}{2} \\ \hline \end{array}
\end{array}
\end{array}$$

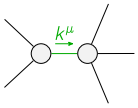
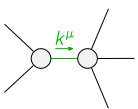
$$\begin{array}{c}
\begin{array}{cc}
\mathcal{J}_{0^+}^{\#1} & \mathcal{J}_{0^+}^{\#1\alpha\beta\chi} \\
\mathcal{J}_{0^+}^{\#1} \dagger & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} \\
\mathcal{J}_{0^+}^{\#1} \dagger^{\alpha\beta\chi} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array}
\end{array}
\begin{array}{cccc}
\mathcal{J}_{1^+}^{\#1\alpha\beta} & \mathcal{J}_{1^+}^{\#2\alpha\beta} & \mathcal{J}_{1^+}^{\#1\alpha} & \mathcal{J}_{1^+}^{\#2\alpha} \\
\mathcal{J}_{1^+}^{\#1} \dagger^{\alpha\beta} & \begin{array}{|c|c|} \hline \frac{2}{3 \text{Det}(1^+)} & \frac{\sqrt{2}}{3 \text{Det}(1^+)} \\ \hline \end{array} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} \\
\mathcal{J}_{1^+}^{\#2} \dagger^{\alpha\beta} & \begin{array}{|c|c|} \hline \frac{\sqrt{2}}{3 \text{Det}(1^+)} & \frac{1}{3 \text{Det}(1^+)} \\ \hline \end{array} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} \\
\mathcal{J}_{1^+}^{\#1} \dagger^\alpha & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} & \begin{array}{|c|c|} \hline \frac{Y_4 k^2}{2 \text{Det}(1^+)} & \frac{Y_3 k^2}{2\sqrt{2} \text{Det}(1^+)} \\ \hline \end{array} & \begin{array}{|c|c|} \hline \frac{Y_3 k^2}{2\sqrt{2} \text{Det}(1^+)} & \frac{Y_2 k^2}{2 \text{Det}(1^+)} \\ \hline \end{array} \\
\mathcal{J}_{1^+}^{\#2} \dagger^\alpha & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} & \begin{array}{|c|c|} \hline \frac{Y_3 k^2}{2\sqrt{2} \text{Det}(1^+)} & \frac{Y_2 k^2}{2 \text{Det}(1^+)} \\ \hline \end{array} & \begin{array}{|c|c|} \hline \mathcal{J}_{2^+}^{\#1\alpha\beta} & \mathcal{J}_{2^+}^{\#1\alpha\beta\chi} \\ \hline \end{array}
\end{array}
\begin{array}{cc}
\mathcal{J}_{2^+}^{\#1} \dagger^{\alpha\beta} & \begin{array}{|c|c|} \hline 0 & 0 \\ \hline \end{array} \\
\mathcal{J}_{2^+}^{\#1} \dagger^{\alpha\beta\chi} & \begin{array}{|c|c|} \hline 0 & \frac{2}{3k^2 \mathcal{K}_{15}^{(4)}} \\ \hline \end{array}
\end{array}
\end{array}$$

Abbreviations used in matrices

$$\begin{aligned}
Y_1 &= 2 \mathcal{K}_{15}^{(4)} - \mathcal{K}_3^{(4)} \& Y_2 = 2 \mathcal{K}_1^{(4)} + 3 \mathcal{K}_{15}^{(4)} \& \\
Y_3 &= 2 \mathcal{K}_1^{(4)} - \mathcal{K}_7^{(4)} \& Y_4 = \mathcal{K}_1^{(4)} - 3 \mathcal{K}_{15}^{(4)} - \mathcal{K}_7^{(4)} \& \text{Det}(1^+) = \frac{3}{2} k^2 (2 \mathcal{K}_{15}^{(4)} - \mathcal{K}_3^{(4)})
\end{aligned}$$

Lagrangian

$$\begin{aligned}
& \mathcal{K}_1^{(4)} \partial_\beta \mathcal{K}_{\chi\delta}^\delta \partial^\chi \mathcal{K}_\alpha^{\alpha\beta} - \mathcal{K}_1^{(4)} \partial_\chi \mathcal{K}_{\beta\delta}^\delta \partial^\chi \mathcal{K}_\alpha^{\alpha\beta} + \mathcal{K}_3^{(4)} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}^\delta + \\
& 6 \mathcal{K}_{15}^{(4)} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta - 2 \mathcal{K}_3^{(4)} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta + 3 \mathcal{K}_{15}^{(4)} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}^\delta - \mathcal{K}_3^{(4)} \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}^\delta + \\
& \mathcal{K}_7^{(4)} \partial^\chi \mathcal{K}_\alpha^{\alpha\beta} \partial_\delta \mathcal{K}_{\chi\beta}^\delta - \frac{1}{2} \mathcal{K}_3^{(4)} \partial_\alpha \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\beta\chi}^\delta + \mathcal{K}_{15}^{(4)} \partial_\delta \mathcal{K}_{\alpha\beta\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi} + \mathcal{K}_{15}^{(4)} \partial_\delta \mathcal{K}_{\beta\alpha\chi} \partial^\delta \mathcal{K}^{\alpha\beta\chi}
\end{aligned}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi} \mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}_{0^+}^{\#1\alpha\beta\chi} == 0$	1	$ \begin{aligned} & \partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\chi} + \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta \partial^\chi \mathcal{J}^{\alpha\beta\delta} + \partial_\delta \partial^\chi \mathcal{J}^{\delta\alpha\beta} + \partial_\delta \partial^\delta \mathcal{J}^{\beta\alpha\chi} = \\ & \partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta \partial^\beta \mathcal{J}^{\alpha\chi\delta} + \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + \partial_\delta \partial^\chi \mathcal{J}^{\beta\alpha\delta} + \partial_\delta \partial^\delta \mathcal{J}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \mathcal{J}^{\chi\alpha\beta} \end{aligned} $	
$\mathcal{J}_{0^+}^{\#1} == 0$	1	$\partial_\beta \mathcal{J}_\alpha^{\alpha\beta} == 0$	
$\mathcal{J}_{1^+}^{\#1\alpha\beta} == \mathcal{J}_{1^+}^{\#2\alpha\beta}$	3	$3 \partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\beta\alpha\chi} + 2 \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\chi\alpha\beta} = 3 \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\alpha\beta\chi}$	
$\mathcal{J}_{2^+}^{\#1\alpha\beta} == 0$	5	$3 \partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 3 \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2 \eta^{\alpha\beta} \partial_\epsilon \partial^\epsilon \partial_\delta \mathcal{J}_\chi^{\chi\delta} = 2 \partial_\delta \partial^\beta \partial^\alpha \mathcal{J}_\chi^{\chi\delta} + 3 (\partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\beta\alpha\chi})$	
Total # constraint(s):	10		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$ \begin{aligned} & (18 \mathcal{K}_1^{(4)} \mathcal{K}_{15}^{(4)2} + 18 \mathcal{K}_{15}^{(4)3} + 18 \mathcal{K}_{15}^{(4)2} \mathcal{K}_3^{(4)} + 6 \mathcal{K}_{15}^{(4)2} \mathcal{K}_7^{(4)} + 6 \mathcal{K}_{15}^{(4)} \mathcal{K}_3^{(4)} \mathcal{K}_7^{(4)} - \mathcal{K}_{15}^{(4)} \mathcal{K}_7^{(4)2} + 2 \mathcal{K}_3^{(4)} \mathcal{K}_7^{(4)2} - \\ & \sqrt{(324 \mathcal{K}_1^{(4)2} \mathcal{K}_{15}^{(4)4} + 8100 \mathcal{K}_{15}^{(4)6} - 1080 \mathcal{K}_{15}^{(4)5} (3 \mathcal{K}_3^{(4)} - 5 \mathcal{K}_7^{(4)}) + 2 \mathcal{K}_{15}^{(4)} \mathcal{K}_3^{(4)} (12 \mathcal{K}_3^{(4)} - 11 \mathcal{K}_7^{(4)}) \mathcal{K}_7^{(4)3} + 4 \mathcal{K}_3^{(4)2} \mathcal{K}_7^{(4)4} + \\ & 36 \mathcal{K}_1^{(4)} \mathcal{K}_{15}^{(4)2} (5 \mathcal{K}_{15}^{(4)} - \mathcal{K}_3^{(4)}) (18 \mathcal{K}_{15}^{(4)2} + 6 \mathcal{K}_{15}^{(4)} \mathcal{K}_7^{(4)} + \mathcal{K}_7^{(4)2}) + 108 \mathcal{K}_{15}^{(4)4} (3 \mathcal{K}_3^{(4)2} - 20 \mathcal{K}_3^{(4)} \mathcal{K}_7^{(4)} + 18 \mathcal{K}_7^{(4)2}) + \\ & 12 \mathcal{K}_{15}^{(4)3} \mathcal{K}_7^{(4)} (18 \mathcal{K}_3^{(4)2} - 72 \mathcal{K}_3^{(4)} \mathcal{K}_7^{(4)} + 29 \mathcal{K}_7^{(4)2}) + \mathcal{K}_{15}^{(4)2} \mathcal{K}_7^{(4)2} (108 \mathcal{K}_3^{(4)2} - 168 \mathcal{K}_3^{(4)} \mathcal{K}_7^{(4)} + 37 \mathcal{K}_7^{(4)2}))) / \\ & (\mathcal{K}_{15}^{(4)} (2 \mathcal{K}_{15}^{(4)} - \mathcal{K}_3^{(4)}) (6 \mathcal{K}_1^{(4)} \mathcal{K}_{15}^{(4)} + 18 \mathcal{K}_{15}^{(4)2} + 6 \mathcal{K}_{15}^{(4)} \mathcal{K}_7^{(4)} + \mathcal{K}_7^{(4)2})) \end{aligned} $
	2	0	$ \begin{aligned} & (18 \mathcal{K}_1^{(4)} \mathcal{K}_{15}^{(4)2} + 18 \mathcal{K}_{15}^{(4)3} + 18 \mathcal{K}_{15}^{(4)2} \mathcal{K}_3^{(4)} + 6 \mathcal{K}_{15}^{(4)2} \mathcal{K}_7^{(4)} + 6 \mathcal{K}_{15}^{(4)} \mathcal{K}_3^{(4)} \mathcal{K}_7^{(4)} - \mathcal{K}_{15}^{(4)} \mathcal{K}_7^{(4)2} + 2 \mathcal{K}_3^{(4)} \mathcal{K}_7^{(4)2} + \\ & \sqrt{(324 \mathcal{K}_1^{(4)2} \mathcal{K}_{15}^{(4)4} + 8100 \mathcal{K}_{15}^{(4)6} - 1080 \mathcal{K}_{15}^{(4)5} (3 \mathcal{K}_3^{(4)} - 5 \mathcal{K}_7^{(4)}) + 2 \mathcal{K}_{15}^{(4)} \mathcal{K}_3^{(4)} (12 \mathcal{K}_3^{(4)} - 11 \mathcal{K}_7^{(4)}) \mathcal{K}_7^{(4)3} + 4 \mathcal{K}_3^{(4)2} \mathcal{K}_7^{(4)4} + \\ & 36 \mathcal{K}_1^{(4)} \mathcal{K}_{15}^{(4)2} (5 \mathcal{K}_{15}^{(4)} - \mathcal{K}_3^{(4)}) (18 \mathcal{K}_{15}^{(4)2} + 6 \mathcal{K}_{15}^{(4)} \mathcal{K}_7^{(4)} + \mathcal{K}_7^{(4)2}) + 108 \mathcal{K}_{15}^{(4)4} (3 \mathcal{K}_3^{(4)2} - 20 \mathcal{K}_3^{(4)} \mathcal{K}_7^{(4)} + 18 \mathcal{K}_7^{(4)2}) + \\ & 12 \mathcal{K}_{15}^{(4)3} \mathcal{K}_7^{(4)} (18 \mathcal{K}_3^{(4)2} - 72 \mathcal{K}_3^{(4)} \mathcal{K}_7^{(4)} + 29 \mathcal{K}_7^{(4)2}) + \mathcal{K}_{15}^{(4)2} \mathcal{K}_7^{(4)2} (108 \mathcal{K}_3^{(4)2} - 168 \mathcal{K}_3^{(4)} \mathcal{K}_7^{(4)} + 37 \mathcal{K}_7^{(4)2}))) / \\ & (\mathcal{K}_{15}^{(4)} (2 \mathcal{K}_{15}^{(4)} - \mathcal{K}_3^{(4)}) (6 \mathcal{K}_1^{(4)} \mathcal{K}_{15}^{(4)} + 18 \mathcal{K}_{15}^{(4)2} + 6 \mathcal{K}_{15}^{(4)} \mathcal{K}_7^{(4)} + \mathcal{K}_7^{(4)2})) \end{aligned} $
Resolved unitarity condition(s):	$ \begin{aligned} & (\mathcal{K}_{15}^{(4)} < 0 \& \mathcal{K}_3^{(4)} < 2 \mathcal{K}_{15}^{(4)} \& \mathcal{K}_1^{(4)} < \frac{-18 \mathcal{K}_{15}^{(4)2} - 6 \mathcal{K}_{15}^{(4)} \mathcal{K}_7^{(4)} - \mathcal{K}_7^{(4)2}}{6 \mathcal{K}_{15}^{(4)}}) \parallel \\ & (\mathcal{K}_{15}^{(4)} > 0 \& \mathcal{K}_3^{(4)} < 2 \mathcal{K}_{15}^{(4)} \& \mathcal{K}_1^{(4)} < \frac{-18 \mathcal{K}_{15}^{(4)2} - 6 \mathcal{K}_{15}^{(4)} \mathcal{K}_7^{(4)} - \mathcal{K}_7^{(4)2}}{6 \mathcal{K}_{15}^{(4)}}) \end{aligned} $		